

MKEC1123 – Advance Microprocessor System

Group Assignment (35%)

This group assignment is designed for a team of **three students** to collaboratively design and develop a **smart adaptive traffic light system** for a **four-way junction**. The primary objective is to implement a system that can **dynamically adjust the green light duration** at each intersection based on real-time **traffic flow**, using the **STM32 Cortex-M4** microcontroller and appropriate sensor technologies. The final output will be a working **physical prototype** that simulates a smart traffic control system capable of adapting to traffic conditions at all four directions of a junction.

The adaptive traffic control system will utilize sensors to detect vehicle presence or count vehicles at each lane. These sensors will serve as inputs to the STM32 microcontroller, which will process the data and determine which lane should receive a green signal, and for how long. The system should be able to **prioritize lanes with higher traffic density**, while also ensuring fairness by preventing any direction from being blocked for an extended period. The prototype will include a **simulation of traffic lights** (Red, Yellow, and Green LEDs) for each of the four directions, controlled based on sensor input and the embedded decision-making logic programmed into the STM32.

As part of the assignment, students are required to conduct research on real-world four-way junction traffic management systems, study the role of adaptive timing in traffic control, and justify their choice of sensors. The team must create a **block diagram** showing how all components are connected and a **flowchart** representing the decision-making algorithm. Programming should be done using STM32CubeIDE or Keil MDK, and the code should be well-commented to highlight key functions and logic. The system must demonstrate the ability to operate in a loop, continuously adapting light durations based on changing traffic conditions simulated through sensor activation.

Each group must submit a detailed **technical report** (7 pages) explaining the background, system design, sensor integration, programming approach, testing process, and results. A set of **pitching slides** (10–15 slides) to potential “customer” must also be prepared for a product pitching, supported by a working **prototype** of the smart traffic light system. Additionally, each student is expected to write a brief **individual reflection** describing their role, contributions, and learning experience throughout the project.

In addition to the report and presentation, each group must submit a **5-minute video** explaining their prototype. This video should clearly demonstrate how the system works, including sensor input simulation, traffic light behavior, and how the STM32 logic adapts based on traffic conditions. The video should also include a brief voiceover or narration explaining the system’s core logic, component functions, and results. This serves both as a demonstration and a form of technical communication to explain the project in a concise and engaging manner.

Hardware Requirements:

In this semester, the **NUCLEO-F446RE** development board or **Blue Pill board** or any similar board (is required with a mini-USB cable (not micro-USB)). The board must be purchased by each group. The board price is approximately RM 60+ and is available from various sites such as Mouser.com, Element14 and digikey.my. You may also get it from your seniors.

Sensors, LEDs and switches.

Minimum Software Requirements

A compiler toolchain is required to get started with the board and for future software development. The suggested software are:

1. STM32CubeIDE
2. STM32CubeMX
3. or Keil MDK

all the above software are freely available.

Milestone and Deliverable

This group project consists of six milestones. The descriptions are as follows;

Milestone	Description	Deliverables	Assessment
1	Familiarizing with the STM32F4 Development Board In the first milestone, you are required to write a simple Blinky application for the STM32F4 board. This is to familiarize yourself with the development board, IDE and verify that the development setup is functioning correctly. The Blinky application is a simple application that simply blinks an LED on the dev board. All your firmware code must be uploaded to a GitHub account.	1. A GitHub account for your code. 2. A video/link to a video showing the Blinky app running on the dev board. 3. A short instruction on the process of setting up the Blinky app (1 page).	1% 2% 2%
2	Proposal of Smart Traffic Light System Write a project proposal following the IEEE conference paper format. The proposal must explain the project background, problem statement, proposed solution using STM32 Cortex-M4, sensor choices, adaptive control strategy, block diagram, traffic control flow charts and references.	1. Proposal report	5%
3	Development Progress (Video Blog) Halfway through the project, record a 5-minute video blog showcasing the current progress. Explain your sensor integration, traffic light logic, any challenges faced, and next steps.	1. Video	5%

4	Pitching and Prototype Demonstration Each group will deliver a live presentation framed as a commercial product pitch explaining the system, followed by a demonstration of the working prototype . The team should clearly show the product's value proposition by demonstrating how the adaptive logic responds to simulated traffic conditions to prioritize higher traffic density while ensuring fairness . The presentation must include the technical design explanation, sensor interfacing, algorithm logic, and observed results, but these elements should be communicated in an engaging, marketable manner to highlight the real-world benefits and commercial viability of the solution	1. Slide presentation (10 slides) 2. Presentation (10-15 minutes)	5%
5	Final Technical Report (Conference Paper Style) Prepare a detailed technical report (7-10 pages) explaining the overall design, implementation, hardware/software methodology, system testing, results, challenges, and conclusion following IEEE conference paper style. The paper consists of the following: 1. Abstract 2. Introduction 3. Problem/Motivation 4. Current Works 5. Methodology (Hardware, Software, Algorithm) 6. Results 7. Discussion 8. Conclusion 9. References * In discussion and conclusion include a brief analysis of your system's scalability, potential manufacturing costs, and how it could be deployed as a commercial solution	1. Technical report (conference paper format). 2. Github (source code/ programming)	10%
6	Prototype Video Students must submit a final 5-minute video showing: - Explanation of the system - Demonstration of traffic adaptation based on sensor input - Overview of development challenges and successes.	1. 5-minute final video submission 2. Q&A during live presentation	5%